04.11.15(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2015

Max. Marks: 100
Class: SE
Name of the Course: Digital Logic Design and Analysis
Course Code: UCEC304

Duration: 3 hrs
Semester: III
Branch: COMP

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q1(a)	Convert $(650.17)_8$ into decimal, binary and hex.	06
Q1(b)	Convert $(110101)_2$ and $(101100)_2$ into gray code.	04
Q1(c)	Generate Odd parity hamming code for binary data: 0111001. Use this hamming code to illustrate how receiver corrects errors.	10
Q2	Design a 2 bit Magnitude Comparator OR Implement a 16:1 Multiplexer using 4:1 Multiplexer	10
Q3 (a)	Minimize using K map and realize using NOR gates only $f(A,B,C,D) = \prod M(4,5,6,7,8,12) \cdot d(1,2,3,9,11,14)$	05
Q3(b)	Simplify using Quine McCluskey Method $f(A,B,C,D,E) = \Sigma\, m(0,1,9,15,24,29,30) + d(8,11,31)$	10
Q3(c)	With the help of a neat diagram, explain the functioning of a 4 bit Bidirectional shit register. OR Explain the operation of a 4 bit Universal Shift Register.	05
Q4(a)	Implement a BCD to Excess-3 code convertor with circuit diagram. OR Design a Modulo- 6 synchronous counter using T flip flop. Un-Used states are Zeros.	10
Q4(b)	Design a 4 bit Ring Counter using D flip flop and explain its working.	10
Q5(a)	Explain the following IC characteristics: Power dissipation, propagation delay, fan in, fan out and voltage levels	10
Q5(b)	Compare TTL and CMOS Logic families based on their IC characteristics.	10
Q6 (a)	Write the entity declaration constructs in VHDL for EXOR gate.	04
Q6 (b)	Write a program in VHDL TO IMPLEMENT 16:1 Multiplexer.	06